

Bangyu Li

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Online portfolio: bangyuli.vercel.app

EDUCATION

Newcastle University

Newcastle upon Tyne, U.K.

MSc Computer Game Engineering

Expected: Sep 2025

Relevant Coursework:

- **Advanced Graphics for Games:** Explored OpenGL-based real-time scene rendering with a focus on the rendering pipeline, mesh and texture, shader programming, and real-time lighting models.
- **Advanced Game Technologies:** Investigated core game engine architecture by implementing robust collision detection, physics constraints, spatial acceleration via quadtree/octree, finite state machine, NavMesh-driven pathfinding, and networking.

University at Buffalo, SUNY

Buffalo, NY, U.S.

BSc Computer Science

Jan 2018 – Aug 2021

Relevant Coursework & Projects:

- **CSE442 - Software Engineering:** Developed a full-stack web application as part of a group project using React and Node.js, *Project Frame*, a Wiki-style community platform for fighting game players.
- **CSE474 – Intro. to Machine Learning:** Built a COMPAS-style risk assessment system using Naïve Bayes and predictive parity. Collaborated on data collection and model evaluation in a team setting.

PROFESSIONAL EXPERIENCE

Breaker Games, Shanghai

Oct 2023 – Apr 2024

UE5 Technical Designer

- Designed and Implemented DLC content for Project Zero 2: Apocalypse Party using Blueprints, including weapons and character talents. The DLC increased player retention by 24% on Steam and received mostly positive reviews.
- Collaborated closely with external translation team and led the localization of PZ2: Apocalypse Party.

Treelab Software, Shanghai

Jun 2022 – Apr 2023

Golang Back-end Engineer

- Developed the workflow execution service using a Temporal + RabbitMQ architecture, improving reliability for specific business scenarios by over 50%.
- Maintained infrastructure functions under a Docker microservice architecture, including logging, i18n (Internationalization) and data synchronization, ensuring stable infrastructure performance.
- Built CI/CD pipelines for automated testing and deployment, significantly accelerating project delivery.

Lilith Games, Shanghai

Jun 2021 – Jun 2022

Technical Designer

- Collaborated with cross-functional teams and oversaw acceptance testing for physics, terrain, and matchmaking systems in Avatar game engine.
- Developed plugin tools that provided versatile technical solutions for the game design and art teams, significantly improving the production pipeline efficiency.
- Designed and implemented multiple gameplay features for Project Party, including collecting mechanics and inventory systems, contributing to a more engaging player experience.
- Reconstructed the Cutscene Editor and designed the quest system for a narrative-driven game framework (Revolver Editor), enhancing the storytelling mechanics and the usability of Avatar engine.

PROJECTS

Phantom Flash – itch.io

A horror puzzle game submitted to NUGDS Winter Jam 2025 and ranked the first among all submissions.

Role: Team Leader, System Designer, Gameplay Programmer. Led a team of five, designed core gameplay loop and puzzles, implemented multiple puzzle mechanics, and developed UI system.

Light CatRun – TapTap | Demo Video: [Youtube](https://www.youtube.com/watch?v=1333333333)

An adorable strategic puzzle adventure. Entered the 2024 TapTap Spotlight GameJam, nominated for 'Best Game', 'Best Theme & Narrative', and 'Best Visual & Audio', and proudly awarded 'Best Visual & Audio'.

Role: UI Programmer, Gameplay Programmer. Designed and implemented core gameplay mechanics, including UI and interactable objects.

Fantasy RPG Demo – [Github](https://github.com) | Demo Video: [Bilibili](https://www.bilibili.com/video/BV1333333333)

A MMORPG demo showcasing my expertise in designing and implementing complex gameplay systems.

- **Networking:** Implemented a C# network messaging framework using Protobuf for real-time, event-driven state synchronization of character movement, ensuring smooth and reliable multiplayer interactions.
- **Game Systems:** Developed core gameplay systems, including inventory, shop, friend list, team-up, and multi-channel chat to enrich multiplayer gameplay experience.
- **User Interface:** Built an intuitive UI framework with UGUI for managing menus, item displays, and inventory interactions, improving the overall user experience and interface responsiveness.

Grappling Hook Demo – [Github](https://github.com)

Recreated the grappling hook mechanics from *Sekiro: Shadows Die Twice* using Unity. Implemented dynamic rope physics and hook detection to create realistic grappling behavior.

SKILLS:

Programming languages:	Proficient in Python / Go / C# / C++ / Lua
Unity Engine:	Proficient in UGUI / Unity plugins / prototyping Experienced with Animator / Rigging / Shader
Unreal 5 Engine:	Proficient in Blueprint scripting / prototyping Experienced with Able Ability System / Behavior tree
Database systems:	Proficient in MySQL / Oracle Database / DBeaver Experienced with MongoDB / SQL Server
Version control tools:	Proficient in Git / Perforce